AMPLENESS IS PRESERVED BY BASE CHANGE

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Ampleness can be checked on an affine open covering of the base. Lemma \ref{lemma-characterize-relatively-ample}, cited at proof lines 8918-8920. Reduces the assertion to the pieces of a covering.
- The morphism between the preimages is affine as a base change of an affine morphism. Lemma \ref{lemma-base-change-affine}, cited at proof lines 8924-8926. Puts the pieces into the affine situation.
- Ampleness is preserved by pullback along an affine morphism. Lemma \ref{lemma-pullback-ample-tensor-relatively-ample}, cited at proof lines 8927-8929. Gives ampleness on each piece, hence globally.

Source reference. Original lemma 0893: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10483–10483. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 01VJ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2448–2448. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 01SD: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2334–2334. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 0892: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10482–10482. Navigation pointer to a statement the

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Lemma 0.2. *Let $f : X \to S$ be a morphism of schemes. Let $\mathcal{L}$ be an invertible $\mathcal{O}_X$-module. Let $S' \to S$ be a morphism of schemes. Let $f' : X' \to S'$ be the base change of f and denote $\mathcal{L}'$ the pullback of $\mathcal{L}$ to X'. If $\mathcal{L}$ is f-ample, then $\mathcal{L}'$ is f'-ample.*

Proof. By Lemma 01VJ it suffices to find an affine open covering $S' = \bigcup U'_i$ such that $\mathcal{L}'$ restricts to an ample invertible sheaf on $(f')^{-1}(U'_i)$ for all i. We may choose U'_i mapping into an affine open $U_i \subset S$. In this case the morphism $(f')^{-1}(U'_i) \to f^{-1}(U_i)$ is affine as a base change of the affine morphism $U'_i \to U_i$ (Lemma 01SD). Thus $\mathcal{L}'|_{(f')^{-1}(U'_i)}$ is ample by Lemma 0892. □